Machine Learning Models – Setting Important Parameters (Code Examples)

This quick guide shows how common machine learning model parameters are set directly in code using scikit-learn and popular libraries.

1. Logistic Regression

```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression(
    C=1.0,
    penalty='l2',
    solver='lbfgs',
    max_iter=200,
    class_weight='balanced'
)

model.fit(X_train, y_train)
```

2. KNN (K Nearest Neighbors)

```
from sklearn.neighbors import KNeighborsClassifier

model = KNeighborsClassifier(
    n_neighbors=5,
    weights='distance',
    metric='euclidean'
)

model.fit(X_train, y_train)
```

3. Decision Tree

```
from sklearn.tree import DecisionTreeClassifier

model = DecisionTreeClassifier(
    criterion='gini',
    max_depth=5,
    min_samples_split=4,
    min_samples_leaf=2,
    max_features='sqrt'
)

model.fit(X_train, y_train)
```

4. Random Forest

```
from sklearn.ensemble import RandomForestClassifier

model = RandomForestClassifier(
    n_estimators=200,
    max_depth=10,
    min_samples_split=4,
    min_samples_leaf=2,
    max_features='sqrt',
    random_state=42
```

```
)

model.fit(X_train, y_train)
```

5. Bagging Classifier

```
from sklearn.ensemble import BaggingClassifier
from sklearn.tree import DecisionTreeClassifier

model = BaggingClassifier(
    estimator=DecisionTreeClassifier(),
    n_estimators=100,
    max_samples=0.8,
    max_features=0.8,
    bootstrap=True
)

model.fit(X_train, y_train)
```

6. AdaBoost

```
from sklearn.ensemble import AdaBoostClassifier

model = AdaBoostClassifier(
    n_estimators=100,
    learning_rate=0.1
)

model.fit(X_train, y_train)
```

7. Gradient Boosting

```
from sklearn.ensemble import GradientBoostingClassifier

model = GradientBoostingClassifier(
    n_estimators=200,
    learning_rate=0.05,
    max_depth=3,
    subsample=0.8
)

model.fit(X_train, y_train)
```

8. XGBoost

```
from xgboost import XGBClassifier

model = XGBClassifier(
    n_estimators=300,
    learning_rate=0.05,
    max_depth=6,
    subsample=0.8,
    colsample_bytree=0.8
)

model.fit(X_train, y_train)
```

9. LightGBM

```
from lightgbm import LGBMClassifier

model = LGBMClassifier(
    n_estimators=300,
    learning_rate=0.05,
    num_leaves=31,
    max_depth=10,
    feature_fraction=0.8
)

model.fit(X_train, y_train)
```

10. CatBoost

```
from catboost import CatBoostClassifier

model = CatBoostClassifier(
    iterations=300,
    learning_rate=0.05,
    depth=6,
    l2_leaf_reg=3
)

model.fit(X_train, y_train)
```